PRATIK SURYA

AI & SOFTWARE ENGINEER | DISTRIBUTED SYSTEMS & GRAPH ML

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Software Engineering (Honours) undergraduate specializing in Artificial Intelligence at Monash University with production experience architecting end-to-end machine learning pipelines, temporal graph neural networks, and scalable cloud microservices. Proven track record deploying autonomous RAG agents, cutting administrative latency by 40%, and maintaining 99.9% infrastructure uptime.

EDUCATION & TECHNICAL EXPERTISE

Monash University | Bachelor of Software Engineering (Honours) — Minor in Artificial Intelligence Aug 2022 – Nov 2026

Relevant Coursework: Data Structures & Algorithms, Advanced Databases, Operating Systems, Intelligent Systems, Cloud Computing, Distributed Software Architecture

Languages & Core:	Python, Java, C++, SQL, JavaScript / TypeScript, PHP, HTML5 / CSS3, Git, Linux / Bash, Docker
AI, ML & Deep Learning:	PyTorch, PyG (PyTorch Geometric), TensorFlow, Scikit-Learn, XGBoost, GraphSAGE, Temporal GNNs, LLMs, LangChain, RAG Vector Retrieval, SHAP, SMOTE
Cloud, Systems & DBs:	AWS Architecture (EC2, S3, IAM), FastAPI, PostgreSQL, MongoDB, MySQL, Oracle, Cisco Networking & DOC Ops, Streamlit

FEATURED AI & SOFTWARE ENGINEERING PROJECTS

FraudLens AI: Adaptive Financial Fraud Investigation System `PyTorch Geometric, GraphSAGE, XGBoost, SHAP, Streamlit`

- Architected an inductive Temporal Graph Neural Network (TGNN) modeling financial transactions as dynamic multigraphs to detect synthetic fraud syndicates.
- Engineered a structural **concept drift detection** algorithm tracking embedding divergence over time to trigger automated retraining before accuracy decay.
- Deployed an interactive investigation dashboard with sub-graph path tracing and SHAP feature attributions for audit transparency.

Enterprise AI Agent & Conversational Booking Automation `Python, LangChain, OpenAI API, FastAPI, PostgreSQL, Docker`

- Designed an autonomous booking agent utilizing natural language intent parsing and entity extraction to automate campus scheduling workflows end-to-end.
- Integrated a **RAG vector retrieval** pipeline querying operational schemas dynamically, cutting administrative request overhead by 40%.
- Containerized asynchronous **FastAPI microservices** with **Docker**, implementing JWT-based **RBAC** and rate-limiting to sustain **sub-100ms** response times.

Deep Sequence Multivariate Time-Series Forecasting Engine `TensorFlow, Keras, LSTM, ARIMA, Scikit-Learn`

- Benchmarked classical ARIMA, ensemble regressors, and deep **bidirectional LSTM** networks across 10 years of multivariate national economic data.
- Implemented rolling-origin cross-validation with lag-feature optimization, boosting directional forecast accuracy by 18%.

Imbalance-Aware Credit Card Fraud Detection Pipeline `Python, Scikit-Learn, SMOTE, K-Means, XGBoost`

- Constructed an anomaly detection pipeline on 283K+ transactions combining **SMOTE** synthetic oversampling with cluster-informed undersampling.
- Optimized cost-sensitive decision boundaries to achieve an **0.8249 F1-Score** and **0.8902 Precision**, mitigating false-positive account freezes.

PROFESSIONAL EXPERIENCE

UNIRAZAK (URIMPACT Department) — *AI & Software Development Intern* Feb 2026 – Apr 2026

- Architected an AI Agent Booking System for university operations, embedding conversational NLP workflows into the enterprise web portal.
- Engineered **RAG vector retrieval** with strict role-based access control (**RBAC**), securing sensitive student records and maintaining data privacy.
- Optimized asynchronous API endpoints for high concurrency, enforcing parameter validation and robust automated error recovery.

Magnifi Machines — *Network & Systems Engineer* Dec 2024 – Feb 2025

- Administered Data Operations Center (DOC) infrastructure across multi-node server clusters to guarantee 99.9% verified uptime.
- Configured Cisco enterprise routers/switches, isolated latency anomalies, and established structured troubleshooting protocols, boosting incident response speed by 20%.

180 Degrees Consulting KL — *Technology Director* Jun 2024 – Dec 2024

- Led 2 developers to architect and launch a central operations portal across 10+ teams, increasing cross-functional workflow speed by 15%.
- Managed web hosting architecture and DNS configurations, maintaining 95% system uptime and cutting cold page response times by 25%.
- Spearheaded responsive UI/UX design revamps, boosting platform engagement by 30% and reducing user drop-off rates by 10%.

The SmartBridge — *Cloud Application Developer* Dec 2022 – Feb 2023

- Evaluated AWS cloud architecture frameworks to identify resource efficiencies, reducing projected infrastructure costs by 15%.
- Built scalable backend microservices improving resource utilization by 20% and delivered **4 functional PoCs** that secured executive approval.

CERTIFICATIONS & RECOGNITION

- **Corizo Letter of Recommendation:** Endorsed by VP of HR for outstanding performance in AI/ML engineering, rapid technical execution, and analytics delivery.
- **Certifications:** AI Engineer Bootcamp (NVIDIA & YTU Startup House Germany); Deep Learning Specialization (DeepLearning.AI); Advanced Neural Networks & NLP (Arithmatrix Tech); AI & ML Specialization (Corizo); AWS Machine Learning Foundations; Walmart Advanced Software Engineering; Java Data Structures & Algorithms; Data Analytics & Visualization.